AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT141
Introduction to Probability and Statistics

Final Examination
Fall Semester 2025/2026

Date: 12 / Jan. /2026

Number of Exam Pages: (3) (including this cover sheet)	**Time Allowed: 2 Hours**

Instructions:

Please read the following instructions before starting:

- Total Number of Questions: 5
- This is a closed book exam.
- Be sure you write your name and ID on the External Answer booklet.
- This exam accounts to 50% of the total mark in the course.
- Budget your time according to the mark assigned to each question.
- Mobile phones and all other mobile communication equipment are NOT allowed.
- Calculators are allowed.

Part 1: MULTIPLE CHOICE QUESTIONS

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Solve all the questions. Each question is worth 2 marks, giving a total of 10 possible marks.

Q–1: If a participant scored 92 points in a math competition where the mean was 80 and the participant's z-score was 2.0, what is the standard deviation of the competition scores?

a) 4
b) 5
c) 8
d) 6

Q–2: For a standard normal distribution, the probability of $z < 0$ is ___

a) 0.5
b) 0
c) -0.5
d) 1

Q–3: In a factory producing smart phones, it's known that the probability of a new phone passing its quality control test is 0.85. If 20 new phones are tested, what's the expected number of phones that will pass the quality test?

a) 15
b) 17
c) 12.5
d) 18

Q–4: If the mean of 10 values is 75, then the sum of the values is ___

a) 750
b) 500
c) 450
d) 700

Q–5: If the mean of a data set is 50 and a value in the set is 65, what is the z-score if the standard deviation is 5?

a) 2
b) 4
c) 1
d) 3

Part 2: ESSAY QUESTIONS

Each question is worth 10 marks. You should answer all FOUR questions, giving a total of 40 possible marks.

Q–1: [3+2+5 Marks]

A bakery produces chocolate muffins, and **5%** of them are known to be **underbaked**. A quality inspector randomly selects **6 muffins** from a batch. The inspector wants to determine the probability that:

I. Exactly **three** underbaked muffins.
II. **No** underbaked muffins.
III. **At least two** underbaked muffins.

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Q-2: [3+3+4 Marks]

A continuous random variable with probability density function $f(x)$ given by

$$f(x) = k\ x^2 \qquad 0 \leq x \leq 2$$ where k is a positive constant.

a) Calculate the value of k.
b) Find P(X < 1).
c) Find P(1< X < 1.5).

Q–3: [5+5 Marks]

a) A quality control engineer at a factory is analyzing the lifespan of spare parts, which follows a normal distribution with a mean of 1000 hours and a standard deviation of 50 hours. She wants to know what percentage of parts are expected to last between 950 hours and 1050 hours. Use this information to calculate the required proportion.

Hint: Use the table below

z	-1	-2.00	1.17	1.18	1.60	2.00	1.80	3.00
P (Z < z)	0.1587	0.0228	0.8790	0.8810	0.9452	0.9772	0.9641	0.9987

b) An electronics company is reviewing the performance of its newly developed light bulbs. A sample of 25 bulbs showed an average lifespan of 1200 hours. Assuming the population standard deviation is known to be 80 hours, construct a 95% confidence interval to estimate the true mean lifespan of the bulbs.

[Given: $Z_{0.025} = 1.96$]

Q–4: [10 Marks]

A food quality inspector is investigating whether the average weight of pasta packages produced by a company has changed from the claimed value of 500 grams. She selects a random sample of 40 packages and finds the sample mean to be 495 grams. Given that the population standard deviation is 10 grams, test the hypothesis at the 0.05 significance level to determine if there is evidence of a difference in the average package weight.

[Given: $Z_{0.025} = 1.96$]

End of questions

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